Logger Hibernate

1. Using Log4j.xml
2. Using Log4j.properties

Jars :- log4j.jar and slf4j.jar

Debugger Level

We have used DEBUG with both the appenders. All the possible options are:

- TRACE
- DEBUG
- INFO
- WARN
- ERROR
- FATAL
- ALL

Appenders in Logger: Apache log4j provides Appender objects which are primarily responsible for printing logging messages to different destinations such as consoles, files, sockets, NT event logs, etc.

Each Appender object has different properties associated with it.

Property	Description
Layout	Appender uses the Layout objects and the conversion pattern associated with them to format the logging information.
Target	The target may be a console, a file, or another item depending on the appender.
Level	The level is required to control the filtration of the log messages.
Threshold	Appender can have a threshold level associated with it independent of the logger level. The Appender ignores any logging messages that have a level lower than the threshold level.
Filter	The Filter objects can analyze logging information beyond level matching and decide whether logging requests should be handled by a particular Appender or ignored.

We can add an Appender object to a Logger by including the following setting in the configuration file with the following method:

```
log4j.logger.[logger-name]=level, appender1,appender..n
```

You can write same configuration in XML format as follows:

```
<logger name="com.apress.logging.log4j" additivity="false">
   <appender-ref ref="appender1"/>
   <appender-ref ref="appender2"/>
</logger>
```

We have used only one appender *FileAppender* in our example above. All the possible appender options are:

- AppenderSkeleton
- AsyncAppender
- ConsoleAppender
- DailyRollingFileAppender
- ExternallyRolledFileAppender
- FileAppender
- JDBCAppender
- JMSAppender
- LF5Appender
- NTEventLogAppender
- NullAppender
- RollingFileAppender
- SMTPAppender
- SocketAppender
- SocketHubAppender
- SyslogAppender
- TelnetAppender
- WriterAppender

We would cover FileAppender in Logging in Files and JDBC Appender would be covered in Logging in Database

Layout

We have used PatternLayout with our appender. All the possible options are:

- DateLayout
- HTMLLayout
- PatternLayout
- SimpleLayout
- XMLLayout

Using HTMLLayout and XMLLayout, you can generate log in HTML and in XML format as well.